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Lema de Schwarz

Lema 1 (de Schwarz). *Sea* $f \in \mathcal{H}(\mathbb{D})$ *tal que* $f(0) = 0 \wedge \forall z \in \mathbb{D} : |f(z)| \leqslant 1$*, entonces* (1) $\forall z \in \mathbb{D} : |f(z)| \leqslant |z| \wedge$ (2) $|f'(0)| \leqslant 1$.

Además, si se cumple la igualdad en **??** *para* $z \neq 0$ *o en* **??**, f *es una rotación.*

***Demostración*:** Sabemos que $f'(0) = \lim_{z\to 0} \frac{f(z)}{z}$ y definimos la función g dada por

$$\forall z \in \mathbb{D} : g(z) = \begin{cases} {}^{f(z)}\!/_{z} & \text{si } z \neq 0 \\ f'(0) & \text{si } z = 0 \end{cases} \implies g \in \mathcal{H}(\mathbb{D}).$$

Sea $r \in (0,1)$, entonces $g \in \mathcal{H}(D(0,r)) \cap \mathcal{C}\left(\overline{D(0,r)}\right)$ y, por el principio del módulo máximo, $\forall z \in \overline{D(0,r)} : |g(z)| < \max_{|w|=r} |g(w)|$.

$$\implies \forall z \in \mathbb{D} : |g(z)| \leqslant \max_{|w|=r} \frac{|f(w)|}{r} \leqslant \frac{1}{r} \xrightarrow{r \to 1^-} 1 \implies |g(z)| \leqslant 1.$$

Por tanto, se verifica que $|g(z)| \leqslant |z|$ y $|f'(0)| = |g(0)| \leqslant 1$.

Por otra parte, si $\exists z_0 \in \mathbb{D}^* : |f(z_0)| = |z_0| \vee |f'(0)| = 1 \implies |g(z_0)| = 1 \vee |g(0)| = 1$. Entonces, $|g|$ alcanza su máximo en el $\mathbb{D}$ y, por el teorema del módulo máximo, $g \equiv C \in \mathbb{C}$ constante con $|C| = 1$ (luego $\exists \theta \in \mathbb{R} : C = e^{i\theta}$). Por tanto, $f(z) = Cz = e^{i\theta} z$. ■

Referenciado en

- `teo-schwarz-pick`
- `teo-formula-aut-disco-unidad`

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